ArmorGalv®

SULFIDE STRESS CRACK TESTING

Element Materials Technology

Document #C02

Element Materials Technology
14805 Yorktown Plaza Dr.
Houston, TX
77040 USA

P 713 692-9151
F 713 696-6307
T 888 786 7555
info.houston@element.com
element.com

Report No. 131195-01

Sulfide Stress Cracking Testing of TDZ Coating Systems
P.O. No. 4512835952

Eva Coronado, Ph.D.
Corrosion Laboratory Manager

December 15, 2020

It is our policy to retain components and sample remnants for a minimum of 30 days from the report date, after which time they may be discarded. The data herein represents only the item(s) tested. This report shall not be reproduced, except in full, without prior permission of Element Materials Technology.

EAR Controlled Data: This document contains technical data whose export and re-export/retransfer is subject to control by the U.S. Department of Commerce under the Export Administration Act and the Export Administration Regulations. The Department of Commerce's prior written approval is required for the export or re-export/retransfer of such technical data to any foreign person, foreign entity or foreign organization whether in the United States or abroad.

These commodities, Technology, or software were exported from the United States in accordance with the Export Administration Regulations. Diversion contrary to U.S. law is prohibited.

This project shall be governed exclusively by the General Terms and Conditions of Sale and Performance of Testing Services by Element Materials Technology. In no event shall Element Materials Technology be liable for any consequential, special or indirect loss or any damages above the cost of the work.

INTRODUCTION

This report presents the results of sulfide stress cracking (SSC) Method A test of TDZ coating systems that you requested, and which was performed by Element Materials Technology (Element). The testing was conducted on 4140 85K yield base metal. Element received 1" OD x 70" LG bar identified as MS-004643-03. Base material MTR documents were provided by the client and they are attached to this report as Appendix A. None of the test results in Appendix A were generated by Element. The coating systems are listed below:

Grade	Coating System Details	Specimen ID
4140	None	545, 552
4140	A / 1.5-2 mil Total TDZ & 1 mil Armorplex / Blue	546 – 548 & 553 – 555
4140	B / 1.5-2 mil Total TDZ & 1 mil Sealer Only / Grey	549 – 551 & 556 – 558

Element was asked to perform SSC tests in accordance with NACE Standard TM0177 Method A using two different test environments. The details are listed below. Test temperature was 75˚F, and test duration was 720 hours.

Environment	Specimen ID	Test Solution	Initial pH	Test Gas	Applied Stress
1	545 – 551	Solution A	2.6 – 2.8	100% H_2S	66% SMYS
2	552 – 558	Solution B	3.4 – 3.6	35% H_2S / CO_2	66% SMYS

SSC METHOD A TEST SPECIMENS

Fourteen (14) standard Method A test specimens were machined from the sample. The specimens were prepared using a low-stress grinder and an automatic longitudinal polisher for the gauge section preparation. A 1000 grit emery cloth was used as the final polishing step. Twelve (12) of the specimens were sent to Atomic Alloys, LLC for coating with two

TDZ coating systems. A representative picture of uncoated and coated specimens is shown in Figures 1 through 3.

Figure 1: Uncoated test specimen before exposure.

Figure 2: Test specimen with Coating A (1.5-2 mil total TDZ & 1 mil Armorplex) before exposure.

ARMORGALV® ARMORPLEX BLUE™

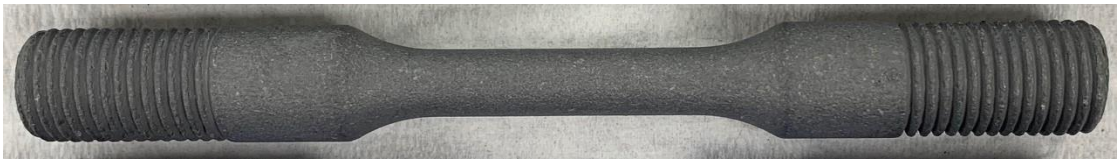

Figure 3: Test specimen with Coating B (1.5-2 mil total TDZ & 1 mil Sealer only) before exposure.

ARMORGALV® - AG3000 -

After machining and/or coating, the specimens were degreased with alkaline detergent cleaner, followed by a two-step solvent rinse with ethanol and acetone, and warm-air dried. Adequacy of degreasing was determined in accordance with ASTM F 21.

LOAD APPLICATION

After degreasing, the specimens were sealed in the test cells. The specimens were stressed to 66 percent of the specified minimum yield strength (SMYS). The SMYS was 85 ksi.

The test cells were then deaerated with nitrogen for minimum 1 hour with a flow rate of 500 mL/min before introducing the test solution.

TEST CONDITION

De-ionized water and reagent grade chemicals were used for the test solution preparation. The test solutions were then deaerated with nitrogen over the weekend with a flow rate of 500 mL/min. The deaerated solution was transferred into each cell, and they were further deaerated with nitrogen for one hour at a flow rate of 500 mL/min. After deaeration, the flow was switched to the test gas at a flow rate of 500 mL/min for one hour and then at a reduced maintenance flow rate for the remainder of the test period (720 hours). The test gas was 100% H_2S (14.7 psi ppH_2S) in the case of environment 1 and 35% H_2S (5 psi ppH_2S) with balance of CO_2 in the case of environment 2.

One specimen per coating system and environment was photographed at approximately 100-hour intervals. The pictures are shown in Figures 4 through 7.

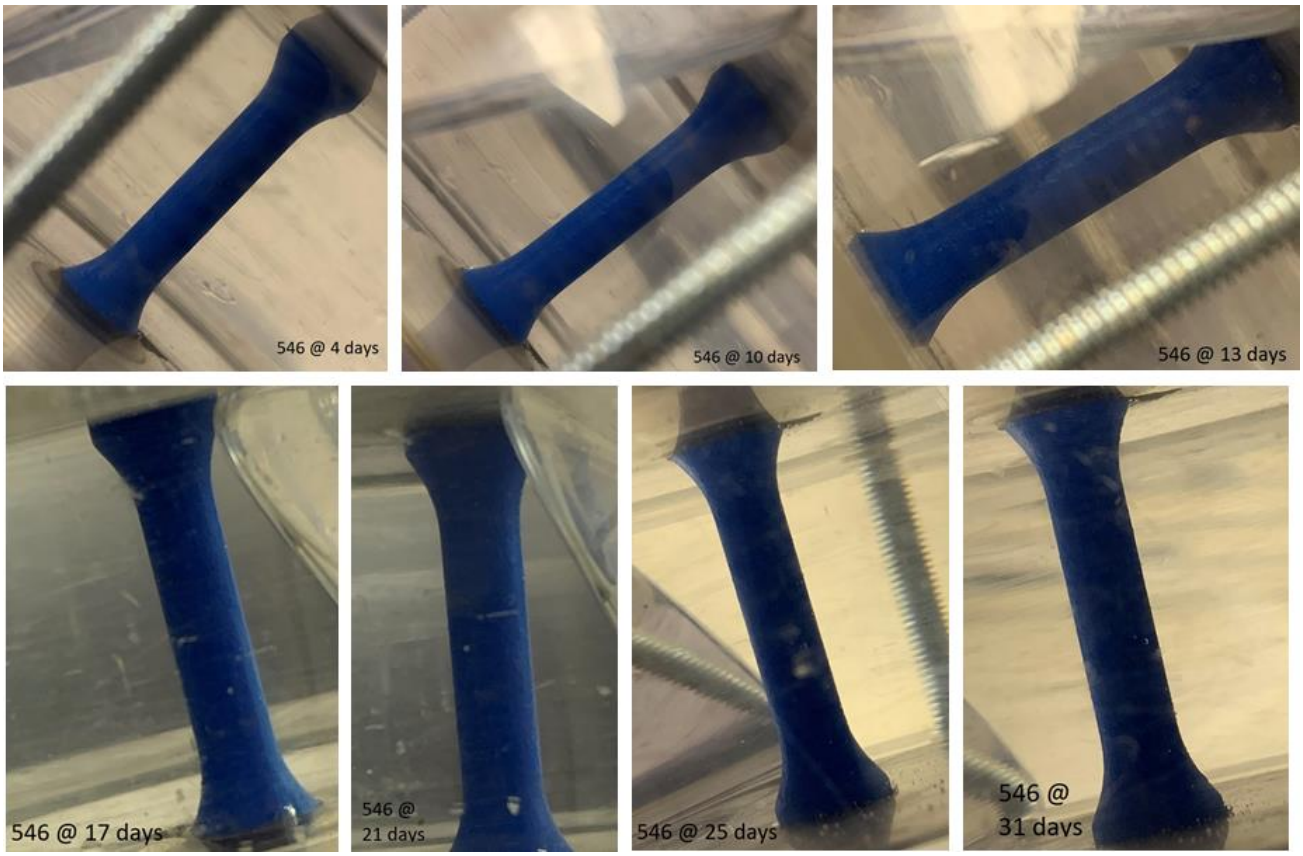

Figure 4: Test specimen 546 with Coating A (1.5-2 mil total TDZ & 1 mil Armorplex) at 100-hour exposure intervals in environment 1.

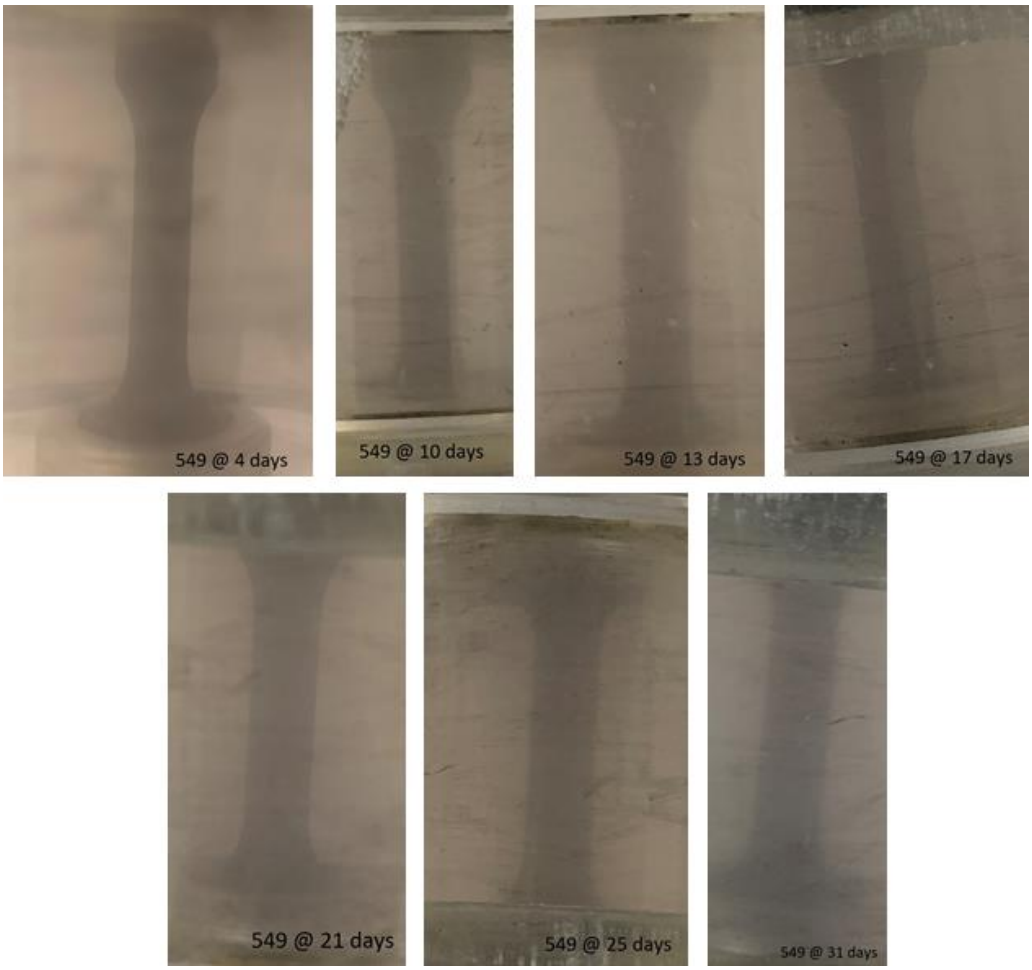

Figure 5: Test specimen 549 with Coating B (1.5-2 mil total TDZ & 1 mil Sealer only) at 100-hour exposure intervals in environment 1.

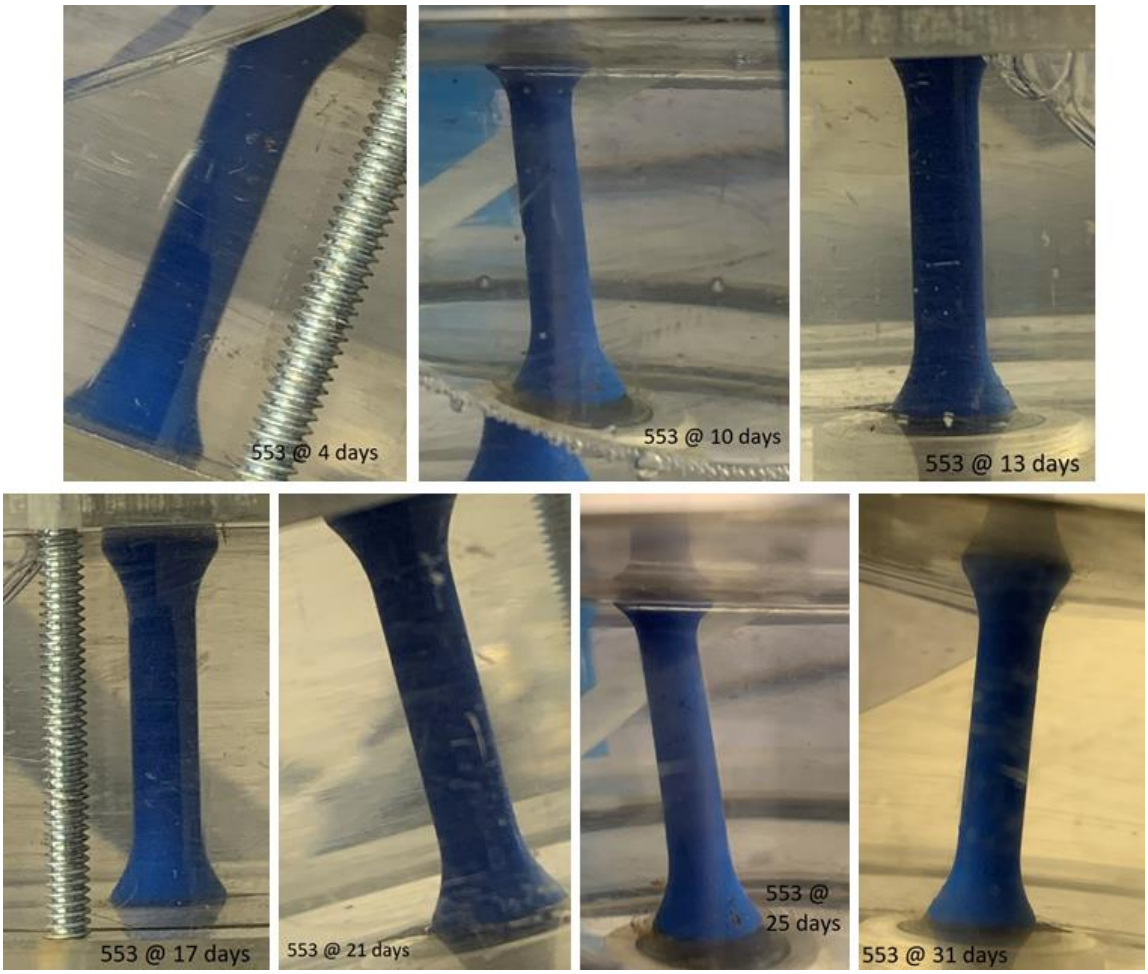

Figure 6: Test specimen 553 with Coating A (1.5-2 mil total TDZ & 1 mil Armorplex) at 100-hour exposure intervals in environment 2.

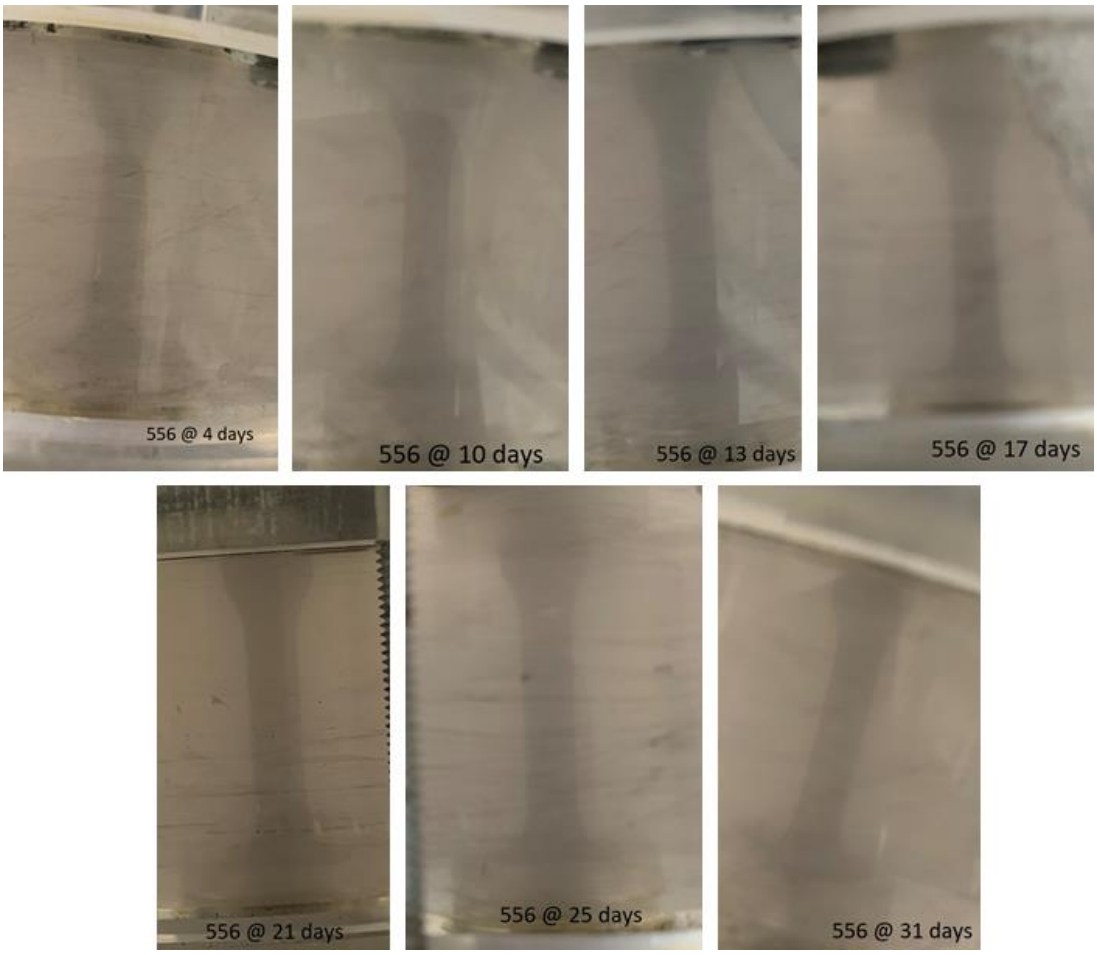

Figure 7: Test specimen 556 with Coating B (1.5-2 mil total TDZ & 1 mil Sealer only) at 100-hour exposure intervals in environment 2.

RESULTS AND CONCLUSIONS

At the end of the test, the final pH from each cell was measured and the H_2S concentration of the last cell was measured. The cells were then purged with nitrogen at a flow rate of 500 mL/min for 30 minutes to remove the H_2S. The initial pH, final pH and H_2S concentration (cH_2S) and test results are listed in Table 1 below. The specimens were then removed from the cells, cleaned and visually evaluated. Figures 8 and 9 show all specimens after the exposure. Figures 10 through 13 show representative detail of reduced section of one specimen per coating system and environment at 20X magnification.

Table 1: Test Results Details

Environment	Specimen ID	Initial pH	Final pH	Final cH_2S (mg/L)	Result
1	545 (Uncoated)	2.6	3.2	2812*	Pass
	546 (Coating A)	2.6	2.6		Pass
	547 (Coating A)	2.6	2.6		Pass
	548 (Coating A)	2.6	2.7		Pass
	549 (Coating B)	2.6	2.7		Pass
	550 (Coating B)	2.6	3.0		Pass
	551 (Coating B)	2.6	2.9		Pass
2	552 (Uncoated)	3.4	3.5	1045**	Pass
	553 (Coating A)	3.4	3.4		Pass
	554 (Coating A)	3.4	3.4		Pass
	555 (Coating A)	3.4	3.4		Pass
	556 (Coating B)	3.4	3.4		Pass
	557 (Coating B)	3.4	3.4		Pass
	558 (Coating B)	3.4	3.4		Pass

Note: * Minimum cH_2S for environment 1 (100% H_2S / 14.7 psi ppH_2S) is 2300 mg/L.

** Minimum cH_2S for environment 2 (35% H_2S balance CO2 / 5 psi ppH_2S) is 805 mg/L.

All specimens passed the test.

Figure 8: Test specimens after the 720-hour exposure in environment 1.

Figure 9: Test specimens after the 720-hour exposure in environment 2.

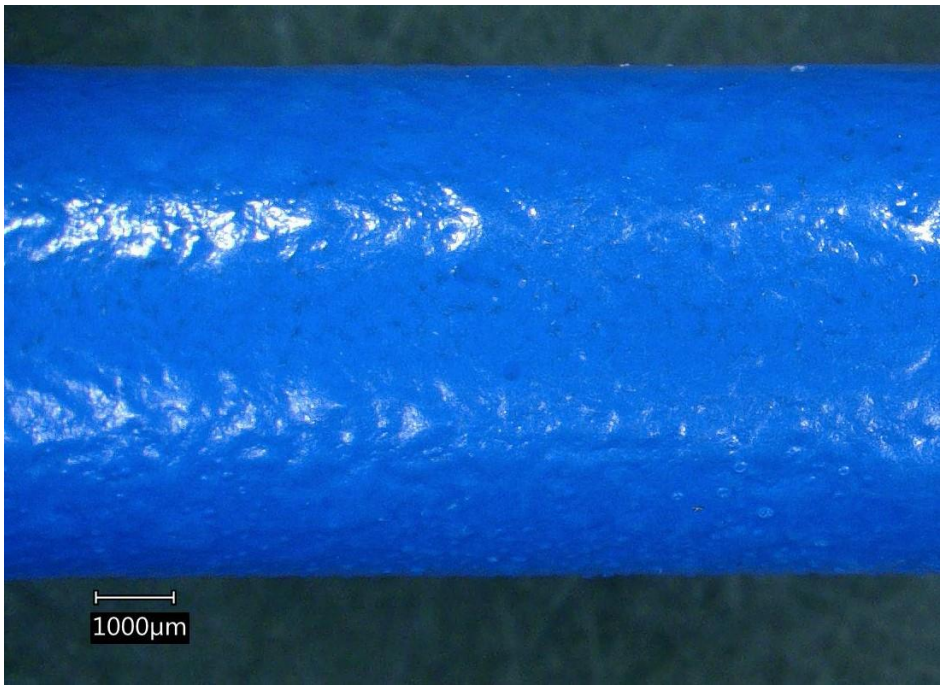

Figure 10: Test specimen 546 with Coating A (1.5-2 mil total TDZ & 1 mil Armorplex) after exposure in environment 1, at 20X magnification.

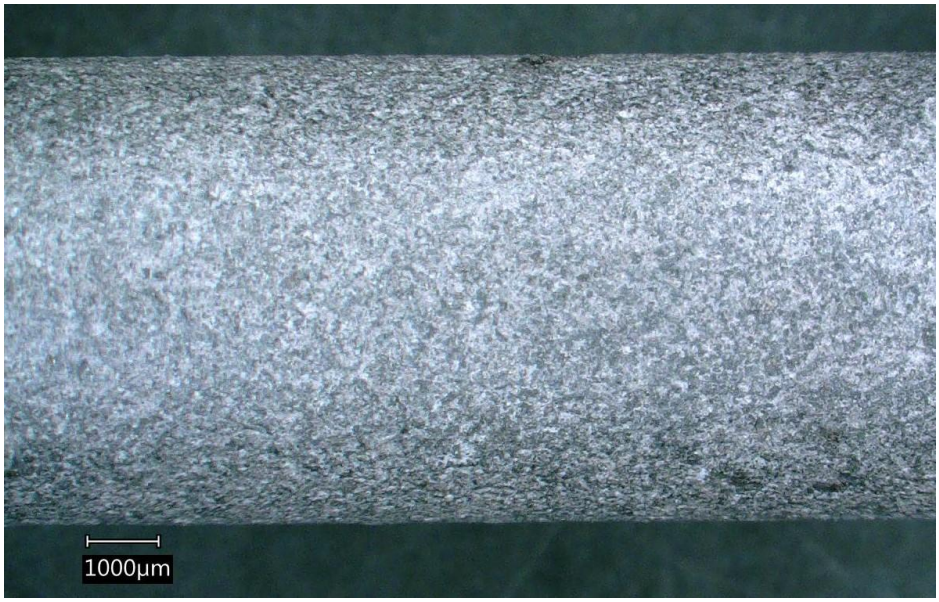

Figure 11: Test specimen 549 with Coating B (1.5-2 mil total TDZ & 1 mil Sealer only) after exposure in environment 1, at 20X magnification.

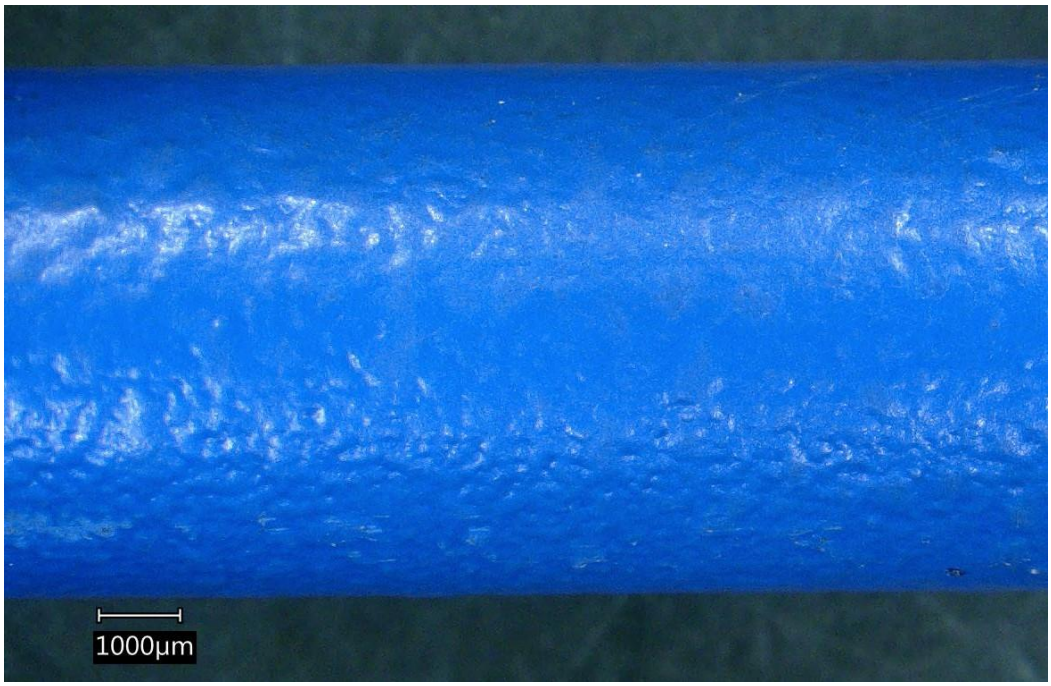

Figure 12: Test specimen 553 with Coating A (1.5-2 mil total TDZ & 1 mil Armorplex) after exposure in environment 2, at 20X magnification.

Figure 13: Test specimen 556 with Coating B (1.5-2 mil total TDZ & 1 mil Sealer only) after exposure in environment 2, at 20X magnification.

Please call us if you have questions about this information or if we may serve you further.

Sincerely,

Eva Coronado, Ph.D.
Corrosion Laboratory Manager

Telephone : (713) 696-6384
E-mail : eva.coronado@element.com

Appendix A

Base material (1” OD x 70” LG bar identified as MS-004643-03) MTR documents.
Note: None of the test results were generated by Element

TURN TECH INCORPORATED

CERTIFICATION PACKAGE FOR

PO# 4512769729-120

THIS CERTIFICATION PACKAGE INCLUDES THE FOLLOWING:

< 9 >	✓	CHEMICAL CERTIFICATION(S)
< 10 >	✓	PHYSICAL CERTIFICATION(S)
< 11 >		HARDNESS CERTIFICATION-WITNESS
< 12 >		FURNACE CHART(S)
< 13 >		STRESS RELIEVE CERTIFICATION
< 14 >	✓	DESCRIPTIVE HEAT TREAT REPORT
< 15 >	✓	CERTIFICATE OF COMPLIANCE
< 16 >		NITRIDE CERTIFICATION
< 17 >		WELD CERTIFICATION(S)
< 18 >	✓	VISUAL EXAMINATION CERTIFICATION(S)
< 19 >		VOLUMETRIC NDE (U/T) CERTIFICATION(S)
< 20 >		SURFACE NDE CERTIFICATION(S) (M/P AND/OR L/P)
< 21 >		SURFACE NDE PRIOR TO WELD CERTIFICATION(S)
< 22 >		POSITIVE MATERIAL INSPECTION (P.M.I.)
< 24 >		COATING CERTIFICATION(S)
< 25 >		PRE-MANUFACTURING MEETING
< 26 >		RADIOGRAPH CERTIFICATION(S)
< 27 >		BOND INTEGRITY/FUSION LINE (U/T) CERTIFICATION(S)
< 28 >		OVERLAY THICKNESS (EDDY CURRENT)
< 29 >		THIRD PARTY INSPECTION WITNESS FORM(S)
< 30 >		SET-OUT SHEET(S)
< 31 >		PROPRIETARY THREAD CERTIFICATION(S)
< 32 >		STRESS/LOAD TEST CERTIFICATION(S)
< 33 >		HYDRO-STATIC TEST CERTIFICATION(S)
< 34 >		RUBBER MOLD CERTIFICATION(S)
< 53 >		PLATING CERTIFICATION

INSPECTED AND PACKAGED BY: Shanna Bry

DATE: 08/21/20

ISO Certified

32007 Industrial Park Dr.
Pinehurst, TX 77362

Phone: (281) 356-1290
E-mail: turn-tech@turn-tech.com

SIGMA TUBE & BAR
The sum of experience, integrity and service.

Certificate of Compliance

Order	01875100
Customer	TURN-TECH INC
Customer PO	97438
Size	1.250 ROUND BAR
Grade	HF 4140 (217-237)
Heat Number	H11805912WX
Lot Number	481385
Specification	MS-4643-03 REV 02
Customer PT#	LG MS-004643-03
Quantity	1 PC 0'-70.50", 1 PC 0'-74"

1) Sigma Tube & Bar LLC hereby certifies that the material described above and on the attached Test Report(s) complies with the terms of the order contract, as presently agreed upon.
2) This order was processed in accordance with Sigma Tube & Bar's Quality Manual Rev F, dated 22 February 2019.
3) No weld repair was performed on this material.

Cody Willeford
Sales Assistant

VNDR: TURN-TECH	VNDR# 10006457
PO# 4512769729-120	CC SER# 01
P/N : 1" OD X 70" LG	MS# 4643-03 REV 02
QP# N/A	HT# H11805912WX-481385
SIGN: *Kristen Troquille*	CTNG: N/A

OL9,10,14(8B) 79077 + 79078

JIANGYIN XING CHENG SPECIAL STEEL WORKS CO., LTD.

NO.297, BINJIANG (E) ROAD, JIANGYIN CITY, JIANGSU PROVINCE, CHINA PC:214429

PRODUCER'S CERTIFICATE OF CHEMICAL AND PHYSICAL ANALYSIS

– EN10204 3.1

DESCRIPTION OF GOODS:HOT ROLLED AND FORGED ROUND STEEL BARS
CONTRACT NO.:XBUSA18303HOU INVOICE NO.:XBUSA18303HOU-2C PO:212278
STEEL IS FREE FROM MERCURY; NO REPAIR WELDING; STEEL IS FREE FROM ANY HARMFUL RADIOACTIVE CONTAMINATION John Deere

Grade	Batch No.	Heat No.	Size(mm)	Size(inch)	Bundles	Pieces	Weight(MT)	Length(Feet)	Condition of Delivery	Manufacture Process	Free from mercury and any harmful radioactive contamination;NO weld repair.
AISI4140QT P11	M1811670	H11805912WX	31.75	1.25	3	190	7.92	22	Q+T	EAF+LF+VD+CC+HR	

	C	Si	Mn	P	S	Cu	Ni	Cr	V	Mo	Al	Ca	H	DI(")	*0001
MIN-	0.40	0.15	0.75	0.000	0.000	0.00	0.00	0.90	0.00	0.15	0.020	0.0000	0.0000	5.5	0.000
MAX-	0.43	0.35	1.00	0.025	0.025	0.25	0.25	1.10	0.05	0.25	0.050	0.0099	0.0003	99.0	0.040
Ladle	0.41	0.26	0.94	0.017	0.004	0.12	0.17	1.05	0.01	0.23	0.026	0.0006	0.0001	6.6	0.021

	Non-Metallic Inclusion											Macro		
	A(T)	A(H)	B(T)	B(H)	C(T)	C(H)	D(T)	D(H)	Ds	S	O	S	R	C
Max	2.5	2	2	1.5	2	1.5	2	1.5				2	2	2
actual	1.0	0.0	0.5	0.5	0.0	0.0	0.5	0.5				1.0	1.0	1.0
actual	1.0	0.0	0.5	0.0	0.0	0.0	0.5	0.0				1.0	1.0	1.0
actual	1.0	0.0	0.5	0.0	0.0	0.0	0.5	0.5						
actual	1.0	0.0	0.5	0.0	0.0	0.0	0.5	0.0						
actual	1.0	0.0	0.5	0.0	0.0	0.0	0.5	0.0						
actual	1.0	0.0	0.5	0.0	0.0	0.0	0.5	0.0						
AVG	1.0	0.0	0.5	0.1	0.0	0.0	0.5	0.2						

Grain Size		Jominy Test (HRC)									Dimension	Visual
		1/16"	1	2	3	4	5	6	7	8		
min	5	min|max	55|60	55|60	54|60	53|59	53|59	52|58		50|57	OK	OK
max	8	actual	58.5	58.5	57.5	57.5	57.5	57.5		57		
actual	6.5		9	10	11	12	13	14	15	16		
		min|max		47|56		44|55		41|54		39|53	UT	Surface
Reduction Ratio		actual		56		55		54		53		
actual	128.8		18	20	22	24	26	28	30	32	OK	OK
		min|max		36|51		34|48		34|46		33|44		
		actual		50		46.5		45		43		

Normalizing			Cooling Media	Austenitizing			Quenching Media	Temp of Quench Media(F)		Tempering			Cooling Method after Tempering	Furnace Hearth Survey &Calibration		Stress Relieving			Furance Type	Heat Treat Lot
Temp.(F)	Tolerance	Time(Min)		Temp.(F)	Tolerance	Time(min)		Starting	Finishing	Temp.(F)	Tolerance	Time(min)		Date	Standard	Temp.(F)	Tolerance	Time(Min)	Continuous	M1811670
				1697	±9	10	Water	77	93.2	1112	±18	330	Air	2017.9.17	AMS-2750					

Mechanical Properties (Longitudinal)

Specimen Location		Yield 0.2% offset ksi	Yield 0.5% EUL ksi	Yield 0.60% ksi	Tensile 4D ksi	YS/TS	Elong 2" %	Reduct of Area %	Specimen Location		Yield 0.2% offset ksi	Yield 0.5% EUL ksi	Yield 0 ksi	Tensile 4D ksi	YS/TS	Elong 2" %	Reduct of Area %
Core	min	110			140		15	45		min							
	max	140			160					max							
actual		135			148	0.912	20	56	actual								
actual		136			148	0.917	21	57	actual								
actual									actual								
actual									actual								
actual									actual								
actual									actual								
actual									actual								

Hardness

Test Location		BHN		Test Location		BHN		Test Location				Test Location			
Surface	min	285		1/2R	min	269			min				min		
	max	341			max				max				max		
actual		313		actual		311		actual				actual			
actual		311		actual		311		actual				actual			
actual				actual				actual				actual			
actual				actual				actual				actual			
actual				actual				actual				actual			
actual				actual				actual				actual			
actual				actual				actual				actual			

VNDR: TURN-TECH
PO# 4512769729-120
P/N : 1" OD X 70" LG
QP# N/A
SIGN: *Kristen Troquille*

VNDR# 10006457
CC SER# 01
MS# 4643-03 REV 02
HT# H11805912WX-481385
CTNG: N/A

TURN-TECH INC PO: 97438, PT# LG MS-004643-03, SP# MS-4643-03 REV 02
SO:01875100 pg 2 of 4 approved by CW 08/04/2020

XINGCHENG STEEL

JIANGYIN XING CHENG SPECIAL STEEL WORKS CO., LTD.

NO.297, BINJIANG (E) ROAD, JIANGYIN CITY, JIANGSU PROVINCE, CHINA PC:214429

PRODUCER'S CERTIFICATE OF CHEMICAL AND PHYSICAL ANALYSIS

-- EN10204 3.1

DESCRIPTION OF GOODS:HOT ROLLED AND FORGED ROUND STEEL BARS
CONTRACT NO.:XBUSA18303HOU INVOICE NO.:XBUSA18303HOU-2C PO:212278
STEEL IS FREE FROM MERCURY; NO REPAIR WELDING; STEEL IS FREE FROM ANY HARMFUL RADIOACTIVE CONTAMINATION John Deere

Grade	Batch No.	Heat No.	Size(mm)	Size(inch)	Bundles	Pieces	Weight(MT)	Length(Feet)	Condition of Delivery	Manufacture Process	Free from mercury and any harmful radioactive contamination;NO weld repair.
AISI4140QT P11	M1811670	H11805912WX	31.75	1.25	3	190	7.92	22	Q+T	EAF+LF+VD+CC+HR	

Energy Charpy Impact Test (10×10×55mm V Notch)

Specimen Location		Long. 77 F	Long. Shear Zone	Long. Lateral Expans	Long. 32 F	Long. Shear Zone	Long. Lateral Expans	Long. 14 F	Long. Shear Zone	Long. Lateral Expans	Long. -20 F	Long. Shear Zone	Long. Lateral Expans	Specimen Location			Shear Zone	Lateral Expan		Shear Zone	Lateral Expans		Shear Zone	Lateral Expans		Shear Zone	Lateral Expans
		ft-lbs	%	mm	ft-lbs	%	mm	ft-lbs	%	mm	ft-lbs	%	mm			ft-lbs	%	mm	ft-lbs	%	mm	ft-lbs	%	mm	ft-lbs	%	mm
Core	min	40			30			1			20				min												
	max										737				max												
actual		77	0.74	75	69	0.68	65	69	0.71	70	59	0.66	65	actual													
actual		74	0.76	70	68	0.70	70	70	0.69	70	65	0.63	60	actual													
actual		76	0.75	75	68	0.71	75	68	0.65	65	61	0.62	60	actual													
actual														actual													
actual														actual													
actual														actual													
actual														actual													

Note	Acc.to ASTM A29-16,A322-13,A304-16,A751-14a,ASTM E381-17,E112-13,E8-16a,E10-15a,E45-13,ASTM A370-17,A255-10(14); UT test acc. to ASTM A388 S1;;Acc. to ASTM E709-15 or TT,ET,MFLT;Ca-Al Treated;Fine grain,fully killed;Nonmetallic inclusion type A-informative purpose;Temperature Control: Contact Thermocouple;Apply to up to 5".ASTM A434-15 Class BD. AISI4140/42;This certificate is written according to EN 10204 3.1 Melted and Manufactured in China.;Tensile specimen:RND.12.5mm;Gage length.2inch;Charpy specimen:10×10×55mm.;;Stress Free. Hot Rolled,Special Quality,Alloy Steel Bars.Steel is free from Mercury contamination,Steel is free from any harmful contamination,No Weld Repairs.;*0001=P+S
Remark	

QUALITY MANGER:Bai Yun

VNDR: TURN-TECH
PO# 4512769729-120
P/N : 1" OD X 70" LG
QP# N/A
SIGN: Kristen Troquille

VNDR# 10006457
CC SER# 01
MS# 4643-03 REV 02
HT# H11805912WX-481385
CTNG: N/A

SHT 10.9 Rev 0

WO# 481385

SPECIALTY HEAT TREAT, INC.

CERTIFICATE OF HEAT TREATMENT

CUST NO:	S196	CUST PH:	281-369-5525
CUST NAME:	SIGMA TUBE & BAR LLC	CUST PO:	S 01875100 01 SR
CUST ADDR:	14315 W. HARDY RD. HOUSTON, TX 77060	CUST HN:	H11805912WX

QUANTITY:	3	MATL:	4140	WGT:	52
LOCATION:	NS-2			STICKER #:	99568
DATE RECD:	7-22-20	DUE DATE:	7-29-20		

DESC:

LT# M1811670

1 1/4"OD X 1 @ 70 1/2"LG; 1 @ 80"LG; 1 TP @ 7"LG

PROCESS INSTRUCTIONS: HEAT TREAT PER MS-004643-03 REV 02

PROCESSED AS FOLLOWS IN A BATCH FURNACE:

MCS: 1 1/4"

CONTROLLED BY FURNACE INSTRUMENT

PROCESS	TIME	TEMP IN DEG F	COOLING METHOD
HARDEN	3 HOURS	1575	QUENCHED TO QUENCH TEMPERATURES
OIL QUENCHED		106-114	
TEMPER	4 HOURS	1290	A/C TO AMBIENT

HARDNESS: CK 6" PROLONGATION (2 PLCS) = 237-237 HBW; CK TP = 237-237 HBW
CK 1 PC(s) = 228 HBW/P

MECHANICAL PROPERTIES OF A 0.503" DIA TP PER ACCU-TEST LAB #: 936235.00 CORE

TENSILE	YIELD .2% OFF	ELONG IN 2"	REDUCTION
111,200 PSI	96,400 PSI	25.4 %	63.2 %

CHARPY IMPACTS 10X10 MM CORE

DIRECTION	TEMPERATURE	FT/LBS	MILS LAT EXP	%SHEAR-%D/F
LONGITUDINAL	-20 °F	97-102-94	70-73-68	100-100-100

HT LOT# 481385

ALL MATERIALS ON THIS ORDER WERE PROCESSED IN ACCORDANCE WITH OUR QUALITY MANUAL SHT 01.2 REV 2

SPECIALTY HEAT TREAT, INC.

By

VNDR: TURN-TECH	VNDR# 10006457
PO# 4512769729-120	CC SER# 01
P/N : 1" OD X 70" LG	MS# 4643-03 REV 02
QP# N/A	HT# H11805912WX-481385
SIGN: Kristen Troguille	CTNG: N/A

TURN-TECH, INC.

32007 INDUSTRIAL PARK DRIVE

PINEHURST, TEXAS 77362

Certificate of Conformance

From: Turn-Tech, Inc

Packing List No: 64189 **Today's Date:** 08/21/20

Shipping Date: 08/21/20 **PO Number:** 4512769729 **Item #:** 120

Part Number: 1" OD X 70" LG MS-004643-03

Material Specification: MS-4643-03 REV 02

Quality Plan: None

Coating Certification: None

Quantity Shipped	Unit	Description		Job Number
1	EA	1" OD X 70" LG MS-004643-03	1" OD X 70" LG MS-004643-03	79077

VNDR: TURN-TECH
PO# 4512769729-120
P/N : 1" OD X 70" LG
QP# N/A
SIGN: Kristen Troquille

VNDR# 10006457
CC SER# 01
MS# 4643-03 REV 02
HT# H11805912WX-481385
CTNG: N/A

We hereby certify that the above mentioned parts meet the requirements of all drawings and specifications listed on the above purchase order, including material, heat treatment, plating and special procedures as applicable.

Clete Jaeger
Authorized Signature
Turn-Tech, Inc

OK 15 79077

Turn-Tech, Inc
32007 INDUSTRIAL PARK DRIVE
PINEHURST, TX 77362
Phone: 281-356-1290
Fax: 281-356-1293

Visual Examination Certification

From: Turn-Tech, Inc
32007 INDUSTRIAL PARK DRIVE
PINEHURST, TX 77362

Report Date: 08/20/20
PO Number: 4512769729 **Item #:** 120
Part Number: 1" OD X 70" LG MS-004643-03

Drawing No: None
Heat No: H11805912WX-481385
Procedure: X-008060 REV 06

Weld Specification: None
Material Specification: MS-4643-03 REV 02
Quality Specification: None
Coating Specification: None

Quantity	Description		Job Number
1 Pcs.	1" OD X 70" LG MS-004643-03	1" OD X 70" LG MS-004643-03	79077

Area(s) Examined:

THIS IS A VISUAL EXAMINATION AFTER FINAL MACHINING FOR SN# 01.

VNDR: TURN-TECH	VNDR# 10006457
PO# 4512769729-120	CC SER# 01
P/N : 1" OD X 70" LG	MS# 4643-03 REV 02
QP# N/A	HT# H11805912WX-481385
SIGN: *Kristen Troguille*	CTNG: N/A

1 Pcs. Examined
1 Pcs. Accepted

Inspected By: Roger E. Houston

TURN TECH INCORPORATED

QUALIFIED INSPECTION METHOD VISUAL EXAMINATION PERSONNEL

ALL TESTING DONE IN ACCORDANCE WITH CAMERON
X-008060 REV 06

ID NUMBER	NAME	QUALIFIED THRU DATES LISTED
<T>2	LORENZO NUNEZ	11/19-11/20
<T>7	SONNY SEBASTIAN	11/19-11/20
<T>9	JORGE GARCIA	11/19-11/20
<T>12	FERNANDO SANCHEZ	11/19-11/20
<T>14	ROBERTO ESPINOZA	11/19-11/20
EMP# 345	CLETE JAEGER	11/19-11/20
<T>10	JUAN LONGORIA	11/19-11/20
<T>15	LUIS MATA	03/20-03/21
<T>17	EDUARDO PENAFIEL	11/19-11/20
EMP# 573	BLAKE URBANOSKY	12/19-12/20

CERTIFIED BY:

JORGE GARCIA — *Jorge Garcia* — DATE Mar 2020

Q.A. MANAGER

VNDR: TURN-TECH
PO# 4512769729-120
P/N : 1" OD X 70" LG
QP# N/A
SIGN: *Kristen Troquille*

VNDR# 10006457
CC SER# 01
MS# 4643-03 REV 02
HT# H11805912WX-481385
CTNG: N/A

ISO Certified

32007 Industrial Park Dr.
Pinehurst, TX 77362

Phone: (281) 356-1290
E-mail: turn-tech@turn-tech.com

Hardness Value Report
Job# 79077
Turn-Tech, Inc.

SHEET 1 OF 1

QUANTITY:	1
MS #:	MS-4643-03 REV 02
HEAT #:	H11805912WX-481385
QP #:	None
DESC:	1" OD X 70" LG MS-004643-03
HARDNESS PERFORMED PER BOM & DWG	

PART #:	1" OD X 70" LG MS-004643-03
MACH DET:	None
PROC:	X-008065 REV 09

MATERIAL HARDNESS RANGE: 217-237 HB

TESTING TYPE (A, B, OR C: HBW

AMBIENT TEMPERATURE: 71°F

EMPLOYEE #: T-14

SERIAL# ↓ ↓ HARDNESS VALUE

1) 229	11)	21)	31)	41)	51)	61)	71)	81)	91)
2)	12)	22)	32)	42)	52)	62)	72)	82)	92)
3)	13)	23)	33)	43)	53)	63)	73)	83)	93)
4)	14)	24)	34)	44)	54)	64)	74)	84)	94)
5)	15)	25)	35)	45)	55)	65)	75)	85)	95)
6)	16)	26)	36)	46)	56)	66)	76)	86)	96)
7)	17)	27)	37)	47)	57)	67)	77)	87)	97)
8)	18)	28)	38)	48)	58)	68)	78)	88)	98)
9)	19)	29)	39)	49)	59)	69)	79)	89)	99)
10)	20)	30)	40)	50)	60)	70)	80)	90)	100)

VNDR: TURN-TECH
PO# 4512769729-120
P/N : 1" OD X 70" LG
QP# N/A
SIGN: Kristen Trequille

VNDR# 10006457
CC SER# 01
MS# 4643-03 REV 02
HT# H11805912WX-481385
CTNG: N/A

NOTES:

REVIEWED BY: Paige E. Thrain

DATE: 8/6/20

TURN-TECH, INC. **CONTROL CHART** **JOB#** 79077

CUST #:	A916
PO# - ITEM#:	4512769729 ITEM#: 120
QUANTITY:	1

PART #:	1" OD X 70" LG MS-004643-03
MACH. DET.:	None
DESC.:	1" OD X 70" LG MS-004643-03
QP#:	None

MS #:	MS-4643-03 REV 02
HEAT #:	H11805912WX-481385
COATING:	None
WELDING:	None

EMPLOYEE #:	537								
SN SER #:	1								
PRINT DIM.	TOLERANCE	PART DIM.	PART DIM.	PART DIM.	PART DIM.	PART DIM.	PART DIM.	PART DIM.	PART DIM.	PART DIM.
1.000	+.005 -.000	1.004								
70.00	LD	70.500								

GAGE# _____

INSPECTOR: Joyce E. Irwin

DATE: 8/20/20

SHEET 1 OF 1

VNDR: TURN-TECH
PO# 4512769729-120
P/N : 1" OD X 70" LG
QP# N/A
SIGN: Kristen Troquille

VNDR# 10006457
CC SER# 01
MS# 4643-03 REV 02
HT# H11805912WX-481385
CTNG: N/A